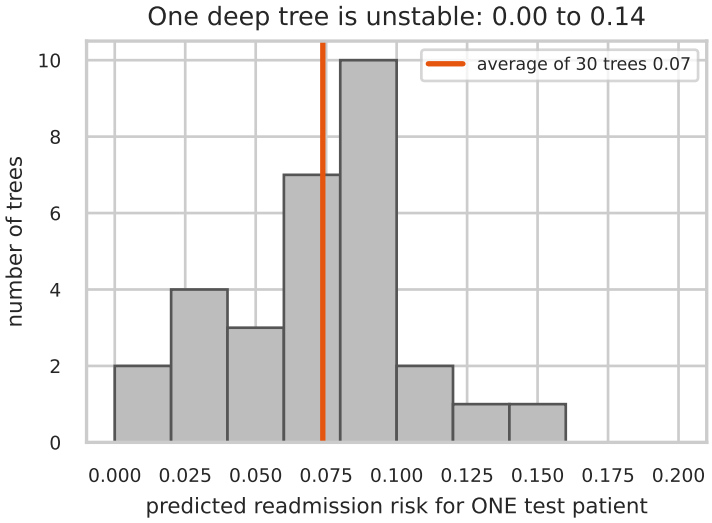
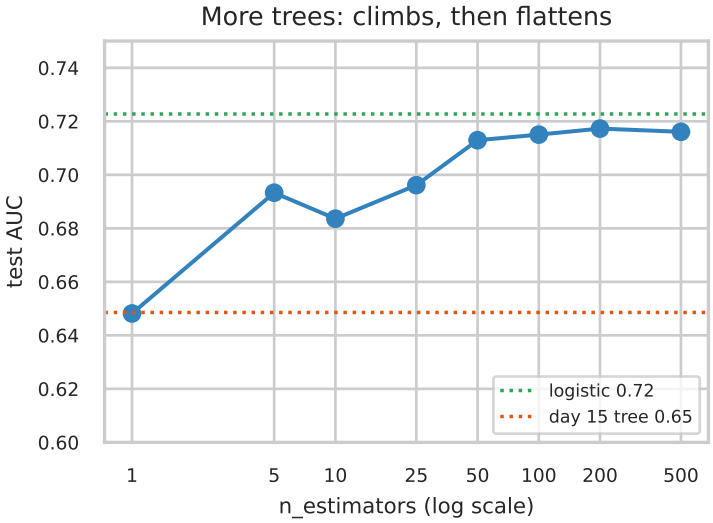
Day 16 Cheat Sheet: Random Forests

Week 3: bagging, random features, out-of-bag scores, and honest importance in scikit-learn

TERM	WHAT IT MEANS	CODE
Bootstrap sample	draw n rows WITH replacement (about 63% unique rows)	rng.integers(0, n, n)
Bagging	fit many models on bootstrap samples, average them	BaggingClassifier()
Random forest	bagged trees + each split sees a random feature subset	RandomForestClassifier()
n_estimators	number of trees; more = steadier, never more overfit	n_estimators=300
max_features	features tried per split; "sqrt" decorrelates trees	max_features="sqrt"
min_samples_leaf	smallest leaf; the main knob against noisy trees	min_samples_leaf=100
Out-of-bag (OOB)	score each row with trees that never saw it	oob_score=True
Impurity importance	how much each feature cut Gini (biased to many values)	rf.feature_importances_
Permutation importance	test AUC drop when one column is shuffled	permutation_importance()
n_jobs	use all CPU cores to fit trees in parallel	n_jobs=-1





THIS DATA (same simulated days 14-15 data, seed 14): 4,000 discharges, 9.0% readmitted, 3,000 train / 1,000 test
30 bootstrap trees (depth 6) gave one patient risks from 0.00 to 0.14: that wobble is what averaging removes
Forest (500 trees, leaf 25): train AUC 0.82, test AUC 0.716, OOB accuracy 0.91 (= base rate, useless), OOB AUC 0.618
CV pick: {'max_features': 'sqrt', 'min_samples_leaf': 100}, CV AUC 0.623, test AUC 0.716
Scoreboard (test AUC): logistic 0.723 | tuned tree 0.649 | random forest 0.716

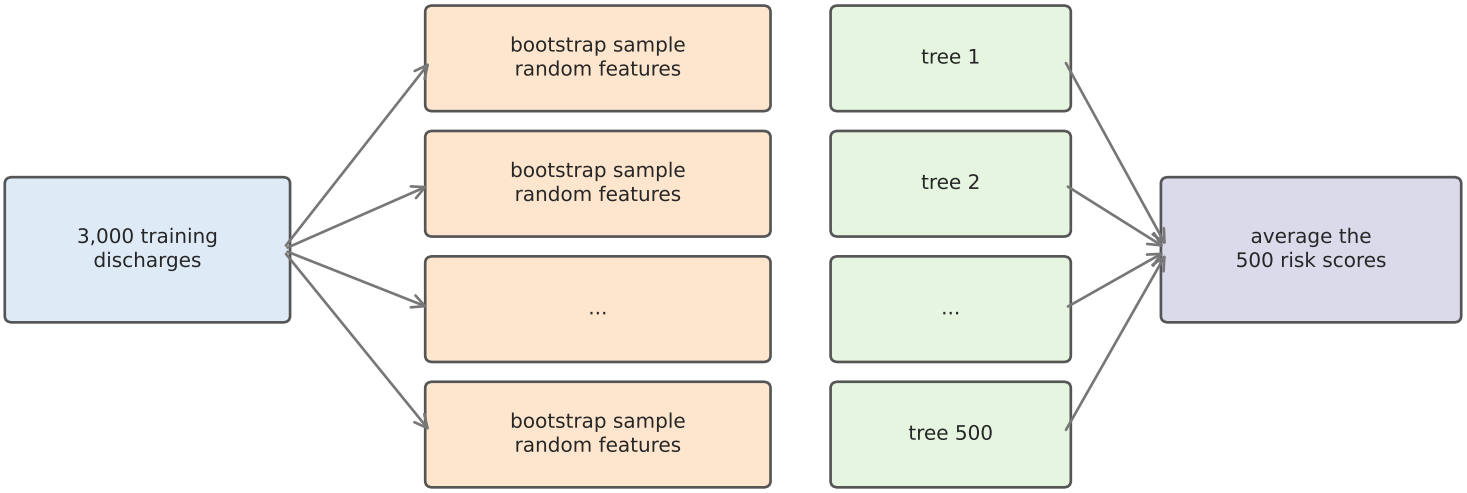
HABITS THAT KEEP YOU HONEST
- Accuracy hides a 9% outcome: OOB accuracy 0.91 equals "nobody gets readmitted". Use AUC
- Tune leaf size and max_features with CV on training data; tree count just needs to be "enough"
- One 1,000-patient test AUC has noise (about +/- 0.03); CV and OOB say about 0.62 here
- Prefer permutation importance on test data; impurity importance favors many-valued columns
- Importance still is not causation: it shows what the model leaned on

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.inspection import permutation_importance
rf = RandomForestClassifier(n_estimators=300, min_samples_leaf=100, max_features="sqrt",
                            oob_score=True, n_jobs=-1, random_state=14).fit(X_train, y_train)
perm = permutation_importance(rf, X_test, y_test, scoring="roc_auc", n_repeats=20)
```

Infographic: 500 Trees Beat One

Simulated readmissions data from days 14-15. Same split, an ensemble model

Bagging + random features = random forest



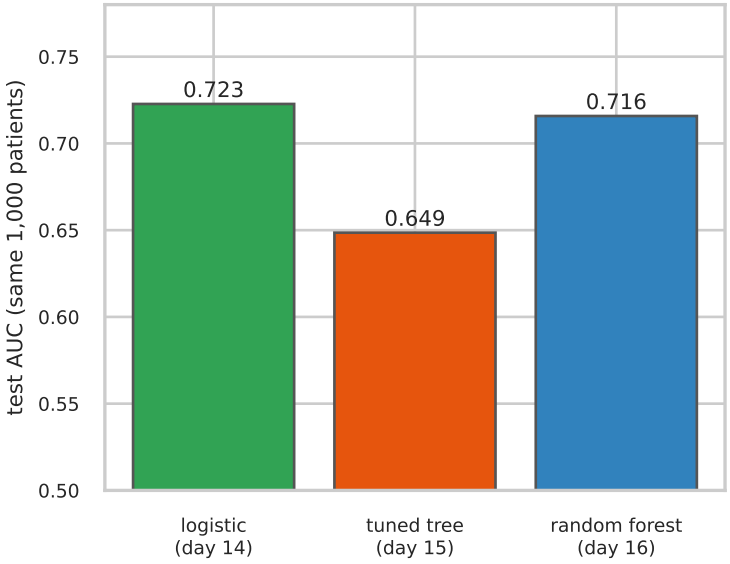
ONE TREE WOBBLES
0.00 to 0.14
for one patient

FOREST AVERAGES
500 trees
test AUC 0.72

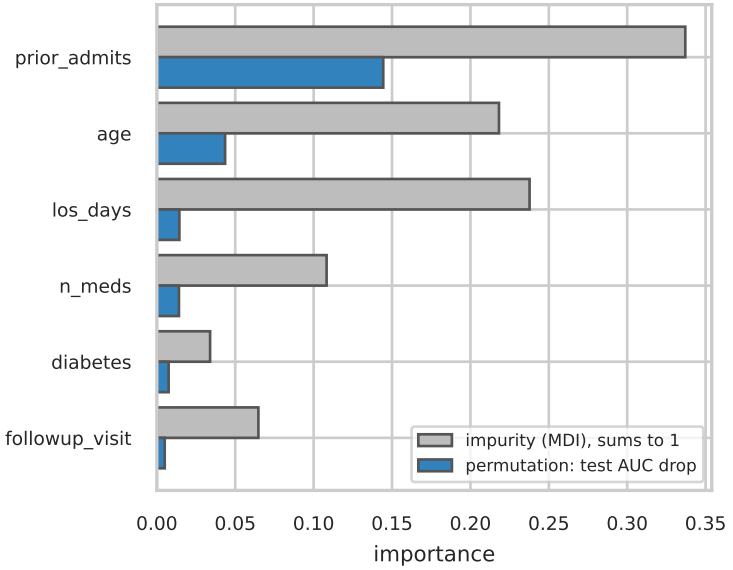
OOB AUC
0.62
no holdout needed

VS LOGISTIC
0.716 vs 0.723
nearly tied

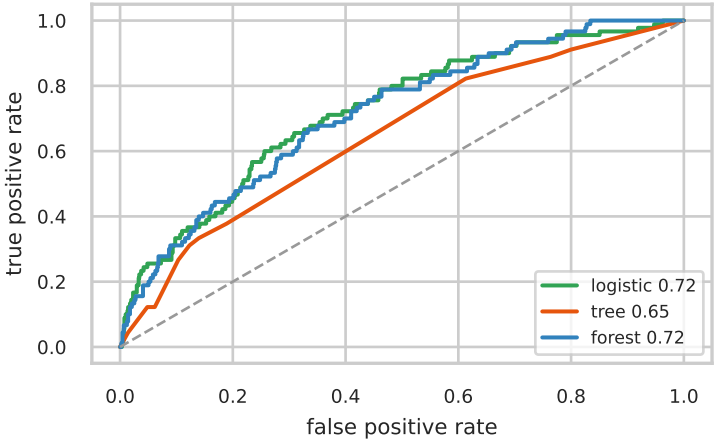
Scoreboard: forest closes the gap



Two importances, two questions



Test ROC curves



Takeaways

- Averaging many unstable trees gave a much better model than one tuned tree (0.65 to 0.72)
- It still only ties logistic regression, because we simulated a smooth, additive risk
- prior_admits dominates both importances; age matters more by permutation than by impurity
- Tomorrow: gradient boosting with XGBoost

Hands-On Worksheet: ds-genai-90-days / day16

60-90 minutes | notebook: day16/random_forest.ipynb + day16/README.md

[ ] mkdir day16, conda activate ds90, open random_forest.ipynb

[ ] Rebuild the days 14-15 data and the same split (snippet 1)

[ ] Bootstrap 30 depth-6 trees; record one patient's risk (snippet 2)

[ ] Histogram the 30 risks; write why averaging helps

[ ] Fit a 500-tree forest with oob_score=True (snippet 3)

[ ] Compare OOB accuracy with the 91% 'nobody readmitted' baseline

[ ] Compute OOB AUC from rf.oob_decision_function_

[ ] Tree-count sweep 1 to 500; line chart of test AUC (snippet 4)

[ ] GridSearchCV on min_samples_leaf x max_features, 5-fold

[ ] Scoreboard: logistic vs tree vs forest test AUC (snippet 5)

[ ] Impurity vs permutation importance side-by-side bar chart

[ ] README: 3 charts, 4 findings, and the test-set noise caveat

[ ] git add day16 && git commit && git push (snippet 6)

[ ] Stretch: write forest risk scores to Snowflake, rank top 10% (snippet 7)

```
from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier(n_estimators=500, min_samples_leaf=25, max_features="sqrt",
                            oob_score=True, n_jobs=-1, random_state=14).fit(X_train, y_train)
oob_auc = roc_auc_score(y_train, rf.oob_decision_function_[:, 1])
p_rf = rf.predict_proba(X_test)[:, 1]
```

One-patient risk range: _____ to _____ OOB accuracy: _____ OOB AUC: _____

Best CV params: _____ CV AUC: _____ Forest / tree / logistic test AUC: ____ / ____ / ____

Day 16 Self-Check Quiz

1. What is a bootstrap sample, and roughly what share of rows appear in it?

   _____

2. A forest has two sources of randomness. Name both.

   _____

3. Why does averaging many trees beat one tree?

   _____

4. OOB accuracy is 0.91. Why is that number almost meaningless here?

   _____

5. Can adding more trees make a forest overfit? Which knob does control it?

   _____

6. Impurity says los_days > age; permutation says age > los_days. Which do you trust?

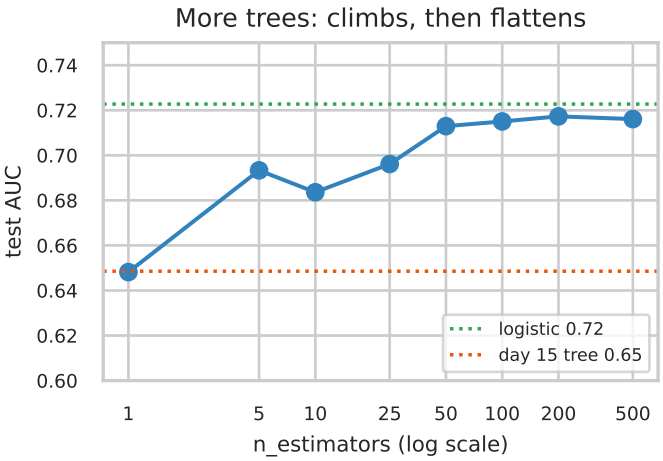
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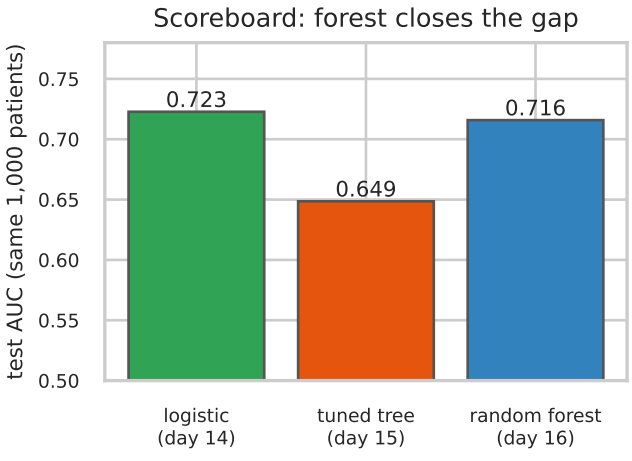
7. CV AUC is 0.62 but test AUC is 0.72. What does that tell you?

   _____

8. The forest ties logistic regression. When would you expect it to win clearly?

   _____





Answers:
1 n rows drawn with replacement from n rows; about 63% are unique, the other ~37% are left out (out-of-bag)
2 each tree gets its own bootstrap sample, and each split only tries a random subset of features (max_features)
3 deep trees have high variance (one patient: 0.00 to 0.14); averaging many partly independent errors cancels much of that noise
4 only 9% are readmitted, so predicting 'no' for everyone is already 91% accurate; use OOB AUC (0.62) instead
5 no, more trees only steadies the average (curve flattens after ~100); min_samples_leaf, max_depth and max_features control overfit
6 permutation: it measures lost test AUC directly; impurity importance favors continuous, many-valued columns like los_days
7 one 1,000-patient test set is noisy; this split looks easy for every model, so compare models on the SAME split and trust CV for tuning
8 when risk has interactions and thresholds (e.g. long stay matters only with prior admits) that a straight-line logit cannot capture